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CUDA:OpenCL translation

Invited Peter Morales Tal Kadosh Ayon Basumallik Niranjan Hasabnis
Tomer Bitan Armand Mousavi Le Chen Matthew Freihofer Sanjna Ravichandar
Inbal Cohen Bolet, Gregory Samyak Jhaveri Erel Kaplan Avraham Evdaev
Yosef Goren Soundararajan, Prema le.chen@anl.gov galoren@technion.ac.il
~~Octavian Udrea~~ ~~Laura Titolo~~

Attachments 📎 CUDA:OpenCL translation 📄 Notes - CUDA:OpenCL translation

Meeting records 📄 Transcript 🎙 Recording

Summary

Erel Kaplan presented the Parcodex app's user-configurable path and highlighted the challenge of high token usage due to passing the entire conversation context between turns, which was confirmed by Tomer Bitan as the agent's operating mechanism; Erel Kaplan, however, successfully reduced some token usage by modifying history usage and removing unnecessary tools, and Le Chen offered to assist with running tests without token constraints. Gal Oren requested that Erel Kaplan conduct specific token usage tests across three GPT versions and different reasoning levels for the upcoming Supercomputing (SC) paper, for which Samyak Jhaveri reported significant progress in defining the HCBench-based benchmark structure, ensuring LLM interaction credibility by strictly limiting the input payload and designing a two-level schema, and Gal Oren instructed Samyak Jhaveri and Erel Kaplan to collaborate closely to integrate this structure into Parcodex within two weeks. Tomer Bitan reported progress on the latent reasoning model, implementing the "realignment matrix" approach, and outlined the plan with Lian Ghayeb to generate the necessary data set for training the Preference Ranking Model (PRM) using the HCBench subset within the next two weeks.

Details

- **Parcodex App and Token Usage** Erel Kaplan is focusing on making the Parcodex path app-like and user-configurable, removing outside dependencies like JSON files or APIs ([00:00:00](#)). They reported a challenge with high token usage due to the entire conversation context being passed from one turn to the next, accumulating to very high total token usage (e.g., 20k tokens at start, 40k+ in the next turn) ([00:01:42](#)). Erel Kaplan noted that most of the tokens were "cached input tokens" from previous messages ([00:05:54](#)).
- **Token Usage Comparison and Testing Requirements** Gal Oren requested Erel Kaplan to conduct a specific test for the paper: selecting 12 kernels and running end-to-end on them using three different GPT versions (5.1, 5.2, 5.3) while also changing the reasoning from "mini" to "extra high" ([00:05:54](#)). This test aims to quantify the hyperparameter choice and address potential reviewer questions about better models or increased reasoning leading to fewer tokens ([00:07:47](#)) ([00:11:17](#)). Erel Kaplan presented initial tests comparing Codex Mini and Codex Max on one kernel, noting that while Mini sometimes used much higher tokens (up to 3 million vs. 1 million), it was still roughly half the price of Max ([00:09:06](#)).
- **Token Usage and System Optimization** Tomer Bitan questioned the payment for the entire context in every prompt, which Erel Kaplan confirmed is how the agent works, where the cost is incurred for the model regenerating and moving the entire context ([00:04:38](#)). Erel Kaplan stated that they attempted to reduce token usage by modifying the history usage in the Codex repo, but removing parts of the context often prevented the task from completing; they managed to save some tokens by removing unnecessary tools and adjusting system instructions ([00:13:42](#)). Erel Kaplan highlighted that saving even 2k tokens in the initial setup could lead to significant overall savings due to the total token accumulation ([00:15:10](#)).
- **Parcodex Application Interface** Erel Kaplan presented the current status of the Parcodex app, which allows users to choose a source directory, select source and target APIs (e.g., CUDA), insert a model, and run multiple jobs simultaneously, showing stages like analysis, translation, optimization, and verification ([00:17:40](#)). The app also includes a feature to analyze performance by comparing the original and translated code ([00:19:30](#)). Tomer Bitan suggested adding a feature to upload tests, ensuring the translated code works for specific user needs ([00:20:31](#)).

- **Supercomputing Paper Strategy** Gal Oren outlined the plan for the Supercomputing (SC) paper submission in less than two months, aiming to showcase the Parcodex system as a product capable of translating from any API to any API. This will involve using a new baseline benchmark based on the collected canonical API-to-API collection of benchmarks, specifically focusing on Samyak Jhaveri's contributions ([00:16:20](#)). Gal Oren emphasized that the paper must include a print screen and access for reviewers to the working tool via a GitHub.io page ([00:19:30](#)).
- **Benchmark Progress and Structure** Samyak Jhaveri reported progress on defining the benchmark structure after meetings, finding 35 repositories suitable for canonical examples, with HCBench being the most relevant, containing 506 kernels and numerous translation pairs ([00:22:50](#)). They developed a pipeline and proposed a two-level schema, including a manifest for discovering canonical translations and a storage structure for files, metadata, and metrics. The Level Two structure, or "spec structure," will hold performance and hardware specifications ([00:23:54](#)) ([00:26:10](#)).
- **Benchmark Design and LLM Interaction** The schema design ensures credibility by controlling what the LLM sees; the LLM only receives the prompt payload (source code and supporting files like headers and make files) for translation, but never the test suites or expected results ([00:27:30](#)) ([00:33:34](#)). Verification involves comparing the final values from the test with expected values derived from a previous run only after the LLM generates and profiles the code, allowing for a defined range of acceptable variance due to non-determinism ([00:28:48](#)). Samyak Jhaveri confirmed they have developed the schema for HCBench and are attempting to automate most of the steps using a CLI interface ([00:30:00](#)) ([00:34:35](#)).
- **Benchmark Integration and Collaboration** Gal Oren instructed Samyak Jhaveri to deliver the HCBench test structure to Erel Kaplan as soon as possible for integration into the new SC paper, serving as the cornerstone for all-to-all API translation ([00:30:57](#)). Gal Oren stressed the need for close collaboration between Erel Kaplan and Samyak Jhaveri to ensure the new benchmark framework can be easily and automatically digested by Parcodex, emphasizing a two-week deadline for this part to be finished for the SC submission ([00:37:36](#)). Niranjana Hasabnis was asked to add Samyak Jhaveri to the Slack or Discord channel for better communication ([00:39:02](#)).

- Agent Memory and Summary Generation** Le Chen inquired about memory control regarding the agent flow, specifically what information is sent back to a previous agent when a verification error occurs ([00:40:01](#)). Erel Kaplan explained that a summary of the session generated by Codex itself, along with the corrupted code, is sent back to the retry mechanism, and this process works well for the agent to succeed on subsequent tries ([00:42:15](#)). Le Chen offered help with running tests without token constraints if Erel Kaplan provides the code ([00:43:35](#)).
- Latent Reasoning Model Progress** Tomer Bitan reported being in phase two of the plan for the reasoning model, focusing on getting the basic main model to run ([00:44:42](#)). They opted to implement the "realignment matrix" approach, described in a retracted paper, as it was the easiest to implement for latent reasoning and comparable in accuracy results ([00:45:50](#)). Tomer Bitan anticipates being done with the latent reasoning model soon, noting that they can use the Technon cluster for running the 70 billion parameter model during this phase and only need a larger machine for Phase 4 ([00:46:59](#)).
- Data Set Generation for PRM Training** Lian Ghayeb is working on adding an LLM as a judge and variable comparison, expecting to be done by Friday. Tomer Bitan detailed the research into training a Preference Ranking Model (PRM), which involves generating a tree of reasoning steps, generating code variations, and then using Lian Ghayeb's scoring mechanism to label each reasoning step based on the final result quality ([00:47:45](#)) ([00:50:04](#)). The plan is to use the HCBench subset to generate the thinking tokens and results needed for this data set, aiming to have a full data set ready by the end of the next week, putting them past Phase 2 in about two weeks ([00:48:47](#)).
- Future Deadlines and Coordination** Gal Oren praised the progress and emphasized the need for deep testing in the following two weeks for the SC paper. They also noted that this research could be suitable for NeurIPS, with an estimated deadline around May. Tomer Bitan pointed out that Lian Ghayeb and a team member were not automatically invited to the meeting, and Niranjana Hasabnis shared the Discord link to resolve communication issues ([00:51:07](#)).

Suggested next steps

- ☐ Samyak Jhaveri will deliver the specific hackbench test to Erel Kaplan as soon as possible.
- ☐ Erel Kaplan will perform a test on 12 kernels using three versions of GPT (5.1, 5.2, and 5.3) and varying reasoning levels (mini to extra high) to quantify the differences in token usage for the paper.
- ☐ Samyak Jhaveri will send the bad looking slide separately to the group.
- ☐ Erel Kaplan will stay in touch with Le Chen about providing the code to test token usage inside the lab.
- ☐ Niranjan Hasabnis will add Samyak Jhaveri to the Slack or Discord channel for better communication.
- ☐ Niranjan Hasabnis will send lian ghrayeb and Tomer Bitan the link to the meeting and Discord channel.
- ☐ Samyak Jhaveri and Erel Kaplan will work closely together to integrate the hackbench structure and framework into the new SC paper and ensure compatibility between the benchmark structure and the paracodex system.
- ☐ The group will keep the chat working and keep everyone updated on their progress.

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